SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 19/20 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I K.Yamini(192572181) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity, relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. RURAL BRANCH OFFICE NETWORK

1. Understanding of the Problem and Constraints

The multinational company is opening a branch office in a remote rural area where high-speed wired Internet is unavailable. The network must support cloud applications, video conferencing and secure communication with headquarters.

The main constraints are:

- No high-speed wired Internet.
- Sufficient bandwidth is required.
- Video conferencing requires low latency and stable connectivity.
- Business operations require high reliability.
- The budget is limited.
- Communication with headquarters must be secure.
- The network should support future expansion.

2. Problem Decomposition

The problem can be divided into:

1. Selecting an Internet connection.
2. Selecting networking devices.
3. Providing security.
4. Providing backup connectivity.
5. Designing a scalable local network.
6. Controlling installation and operating costs.

3. Application of Constraints

Constraint	Solution
No wired broadband	4G/5G fixed wireless
Limited budget	Wireless connection instead of expensive leased line
Video conferencing	5G with sufficient bandwidth and QoS

Uninterrupted operation	Dual-WAN with backup connection
Secure HQ communication	Site-to-site VPN
Network protection	Firewall
Future expansion	Managed switch and scalable access points
Power failures	UPS

4. Recommended Network Design

The recommended architecture is:

4G/5G Network → 5G Router → Firewall → Managed Switch → Computers / Servers / Wi-Fi Access Points

A secondary satellite or wireless connection can be connected to the firewall as a backup WAN.

The main networking devices are:

- 4G/5G router
- Dual-WAN router/firewall
- Managed Ethernet switch
- Wireless access points
- VPN gateway
- UPS

5. Logical Reasoning and Strategy

A 4G/5G connection is suitable because wired high-speed Internet is unavailable. 5G provides better bandwidth and lower latency where coverage is available.

However, a single connection may fail because of network outages or signal problems.

Therefore, a backup wireless or satellite connection should be provided.

The firewall can automatically perform failover when the primary connection fails.

A site-to-site VPN encrypts communication between the branch office and headquarters.

6. Accuracy, Feasibility and Innovation

This solution is practical because it avoids expensive wired infrastructure while providing reliable connectivity. The use of dual-WAN automatic failover is an efficient approach for uninterrupted business operations.

7. Conclusion

Therefore, the best solution is a 5G-based primary connection with a backup wireless/satellite connection, firewall, managed switch, Wi-Fi access points, UPS and site-to-site VPN. This design balances bandwidth, reliability, security, cost and scalability.

Rubric Coverage

Rubric	Marks
Understanding of Problem & Constraints	2
Problem Decomposition	1.5
Application of Constraints	2
Logical Reasoning & Strategy	1.5
Solution Accuracy & Feasibility	1.5
Creativity & Innovation	0.5
Clarity of Presentation	1
Total	10

2. UNIVERSITY CAMPUS WIRELESS NETWORK

1. Understanding of Problem and Constraints

The university needs wireless Internet for more than 3,000 students across classrooms, laboratories, libraries and hostels.

The major constraints are:

- Large number of simultaneous users.

- Campus-wide coverage.
- Limited number of access points.
- Signal interference.
- Secure user access.
- Limited budget.
- Requirement for future expansion.

2. Problem Decomposition

The problem can be divided into:

1. Access point placement.
2. Frequency band selection.
3. Interference reduction.
4. User capacity management.
5. Authentication.
6. Network segmentation.
7. Cost optimization.
8. Future scalability.

3. Application of Constraints

Constraint	Solution
3,000+ students	High-density Wi-Fi 6
Simultaneous users	Load balancing and airtime fairness
Limited APs	User-density-based AP placement
Interference	Channel planning
High performance	5 GHz as primary band
Coverage	2.4 GHz where required
Security	WPA3-Enterprise and 802.1X
User separation	VLANs
Limited budget	Strategic AP deployment
Future growth	Scalable wireless infrastructure

4. Proposed Network Architecture

```
        Internet
           |
         Router
           |
        Firewall
           |
      Core Switch
           |
  Wireless Controller
     /     |     \
   AP     AP     AP
   |      |      |
Students Faculty Guests
```

5. Access Point Placement

Access points should be placed based on user density, not simply equal spacing.

More AP capacity should be provided in:

- Classrooms
- Laboratories
- Libraries
- Hostels
- Seminar halls

AP placement should consider walls, building structure, expected users and interference.

6. Frequency Band Selection

2.4 GHz:

Provides greater range and better wall penetration but has greater interference.

5 GHz:

Should be the primary band because it provides higher performance and more channels for high-density environments.

6 GHz:

Can be introduced for compatible devices when budget permits.

7. Logical Reasoning and Interference Management

Since the number of APs is limited, placing them randomly would result in poor performance. Therefore, high-density locations should receive priority.

Interference can be reduced through:

- Proper channel assignment.
- Transmit-power control.
- Avoiding excessive AP overlap.
- Band steering.
- Centralized wireless management.

Load balancing distributes users across available APs.

8. Authentication and Security

The recommended authentication method is:

WPA3-Enterprise + 802.1X + RADIUS

Separate SSIDs and VLANs can be used for:

- Students
- Faculty
- Guests
- Administration
- IoT devices

This prevents unauthorized users from accessing sensitive university resources.

9. Cost and Feasibility

The university can control cost by installing APs primarily in high-density areas instead of deploying excessive APs everywhere.

Existing suitable switches can be reused, and PoE switches can provide both network connectivity and power to APs.

10. Innovation and Future Scalability

The use of Wi-Fi 6, centralized management, load balancing, band steering and user-density-based AP placement provides an efficient solution with limited hardware.

The network can later be expanded by adding APs, increasing switch capacity and supporting newer wireless technologies.

11. Conclusion

A centrally managed Wi-Fi 6 network with strategic AP placement, 5 GHz priority, WPA3-Enterprise, 802.1X authentication, VLANs and interference management provides a balanced solution.

It satisfies the requirements of coverage, performance, security, cost and scalability.

3. SECURE NETWORK ARCHITECTURE FOR A FINANCIAL INSTITUTION

1. Understanding of Problem and Constraints

A financial institution handles highly sensitive customer and financial information. The network must provide strong cybersecurity while allowing authorized employees to access required services.

The major constraints are:

- Sensitive customer information.
- Unauthorized access must be prevented.
- Strict security policies.
- Compliance requirements.
- High network performance.
- Controlled operational cost.
- Fast detection of attacks.
- Continuous availability of banking services.

2. Problem Decomposition

The problem can be divided into:

1. Network segmentation.
2. Firewall deployment.
3. Authentication and authorization.
4. IDS/IPS deployment.
5. Secure remote access.
6. Customer database protection.
7. Monitoring and logging.
8. Performance and cost optimization.

3. Application of Constraints

Constraint	**Solution**
Sensitive customer data	Isolated database VLAN
External attacks	Perimeter firewall
Internal threats	Internal firewall
Public services	DMZ
Unauthorized access	MFA and RBAC
Network attacks	IDS/IPS
Remote access	Secure VPN
Compliance	Logging and auditing
Performance	QoS and optimized firewall rules
Budget	VLAN-based segmentation and centralized management

4. Proposed Secure Network Architecture

```
        Internet
           |
      Edge Router
           |
   Perimeter Firewall
           |
          DMZ
       /       \
 Web Server   Public Services
           |
   Internal Firewall
           |
      Core Switch
    /      |      \
Employee  Server   Database
```

VLAN VLAN VLAN

5. Network Segmentation

VLANs should separate different users and systems.

VLAN	Purpose
VLAN 10	Employees
VLAN 20	Finance/Admin
VLAN 30	Customer Database
VLAN 40	Application Servers
VLAN 50	Network Management
VLAN 60	Guests

The customer database should be placed in a highly restricted network.
Only authorized application servers and users should be able to communicate with it.

6. Firewall Deployment

A perimeter firewall should protect the organization from external threats.
An internal firewall should control communication between sensitive internal networks.
Firewall rules should follow the principle:
Allow only required traffic and deny unauthorized traffic.
This reduces the attack surface.

7. Authentication and Authorization

The organization should use:

- Multi-factor authentication.
- Strong passwords.
- Role-Based Access Control.
- 802.1X authentication.
- Least-privilege access.

For example, employees should only access customer information required for their specific job.

8. IDS/IPS and Monitoring

An IDS/IPS should monitor network traffic and detect:

- Port scanning.
- Malware.
- Unauthorized access.
- Suspicious connections.
- Abnormal network behavior.

The IPS can automatically block known malicious traffic.
Security logs should be collected and regularly reviewed.

9. Secure Remote Access

Remote employees should connect through a secure VPN.
VPN access should use:

- Encryption.
- Multi-factor authentication.
- Device verification.
- Role-based authorization.

Direct exposure of internal systems to the Internet should be avoided.

10. Performance, Compliance and Budget

Security should not unnecessarily reduce network performance.
Performance can be maintained through:

- Properly sized firewalls.
- Efficient firewall rules.
- VLAN segmentation.

- QoS.
- Continuous monitoring.

Costs can be controlled through:

- Centralized security management.
- Automated monitoring.
- Reuse of suitable existing switches.
- Logical VLAN segmentation.
- Scalable security devices.

For compliance, the organization should maintain:

- Access logs.
- Security event logs.
- Audit records.
- Regular security assessments.
- Controlled administrative access.

11. Creativity and Innovation

The proposed defense-in-depth architecture combines multiple security layers:
Firewall → DMZ → Internal Firewall → VLAN → MFA → IDS/IPS → Encryption → Monitoring

If one security layer fails, the remaining layers provide additional protection.

Using VLANs provides strong logical separation without requiring completely separate physical networks, helping reduce cost.

12. Conclusion

The financial institution should implement a segmented defense-in-depth network using firewalls, DMZ, VLANs, MFA, RBAC, VPN, IDS/IPS, encryption and continuous monitoring. This design provides a practical balance between security, performance, compliance and budget, while protecting sensitive customer information and maintaining reliable banking operations.